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problem 1

(a) given regression model, $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$

It is TRUE that the estimates of $\hat{\beta}$ in above regression model differ for two different samples. This happens because β_0 and β_1 depend on input (x)

$$\beta_0 = \bar{y} - \beta_1 \bar{x} \quad ; \quad \beta_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x}_i) y_i}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

(b)

$$\sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \le \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i)^2$$

TRUE, $\hat{\beta}_0$ and $\hat{\beta}_1$ minimize sum of the square residuals $\left(\sum_{i=1}^{n} (y_i - f(x_i))^2\right)$. So, they are always less than square residuals of even true parameters β_0 and β_1.

(c)

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad (\text{L.R of } Y \text{ on } x)$$

$$Y_i = \gamma_0 + \gamma_1 X_i + \gamma_2 Z_i + \delta_i \qquad (\text{L.R of } Y \text{ on } X \,\&\, Z)$$

So, if a variable Z_i is omitted in Y_i, then β_1 is biased. Because of this the values of x can be changed to accomodate it, hence the statement is FALSE

(d) LR of Y on X, $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$

LR of Y on X & Z, $Y_i = \gamma_0 + \gamma_1 X_i + \gamma_2 Z_i + \delta_i$

$$\sum_{i=1}^{n} (Y_i - \hat{\gamma}_0 - \hat{\gamma}_1 X_i - \hat{\gamma}_2 Z_i)^2 < \sum_{i=1}^{n} (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

TRUE, as additional variable (Z_i) is include in the model, this will improve fit which means less sum of squares.

(e) FALSE, unadjusted R^2 will tend to increase as we add more variables, even if they're not relevant, hence unadjusted R^2 is not sufficient.

problem 2

$$e = Y - \hat{\beta}_0 - \hat{\beta}_1 X$$

$$Y = \hat{\beta}_0 + \hat{\beta}_1 X$$

(a) plot 4.

The points dont follow the linear (showing linear relationship), this means the model is unable to capture the variability.

So residual will have a spread around zero without any clear pattern.

(b) Plot 1

This has strong relationship (linear) so the residual plots won't have much scattering

as X increases Y decreases

similar to Y vs X plots

(c) plot 2

here, for different values of X, Y tends to be in a range with a parallel fit.

So the estimate will have similar

(d) plot 3.

In Y vs X, there is a pattern of how the points are spread around the linear model, the estimate will also have similar spread but around a line parallel to X axis